DSP48A1 Behavioral Model — Design Report

Top module: DSP | Target part: xc7a200tffg1156-3 | Clock: 100 MHz on W5

1. RTL Code

1.1 DSP.v

```
module DSP (A, B, D, C, BCIN, CLK, CARRYIN, OPMODE, RSTA, RSTB, RSTM, RSTP, RSTC, RSTD, RSTCARRYIN,
RSTOPMODE, CEA, CEB, CEM, CEP, CEC, CED, CECARRYIN, CEOPMODE, PCIN, BCOUT, PCOUT, P, M, CARRYOUT,
CARRYOUTF) ;

parameter A0REG = 0,
          A1REG = 1,
          B0REG = 0,
          B1REG = 1,
          CREG = 1,
          DREG = 1,
          MREG = 1,
          PREG = 1,
          CARRYINREG = 1,
          CARRYOUTREG = 1,
          OPMODEREG = 1,
          B_INPUT = "DIRECT",
          CARRYINSEL = "OPMODE5",
          RSTTYPE = "SYNC" ;


input  [17:0] A , B , D , BCIN ;
input  [47:0] C ;
input  [7:0] OPMODE ;
input  CLK, CARRYIN, RSTA, RSTB, RSTM, RSTP, RSTC, RSTD, RSTCARRYIN, RSTOPMODE, CEA, CEB, CEM, CEP, CEC,
CED, CECARRYIN, CEOPMODE, PCIN ;

output [17:0] BCOUT ;
output [47:0] PCOUT , P ;
output [35:0] M ;
output CARRYOUT , CARRYOUTF ;

wire cin , CYI ;
wire [17:0] A0_reg , B0_reg , B_in , D_reg , pre_addsub_out , B1 , B1_reg , A1_reg ;
wire [47:0] C_reg , Z_out , X_out ;
wire [48:0] Post_addsub_out ;
wire [35:0] multiplier_out , M_reg ;

assign B_in = B_INPUT == "DIRECT" ? B : B_INPUT == "CASCADE" ? BCIN : 18'b0 ;

pip_mux  #( .n(18) , .PIPLINING(A0REG) , .RST_TYPE(RSTTYPE) )  A0_inst ( A , RSTA , CLK , CEA , A0_reg );
pip_mux  #( .n(18) , .PIPLINING(B0REG) , .RST_TYPE(RSTTYPE) )  B0_inst ( B_in , RSTB , CLK , CEB , B0_reg
);
pip_mux  #( .n(18) , .PIPLINING(DREG)  , .RST_TYPE(RSTTYPE) )  D_inst ( D , RSTD , CLK , CED , D_reg );
pip_mux  #( .n(48) , .PIPLINING(CREG)  , .RST_TYPE(RSTTYPE) )  C_inst ( C , RSTC , CLK , CEC , C_reg );

assign pre_addsub_out = OPMODE[6] ? D_reg - B0_reg : D_reg + B0_reg ;
assign B1 = OPMODE[4] ? pre_addsub_out : B0_reg ;

pip_mux  #( .n(18) , .PIPLINING(B1REG) , .RST_TYPE(RSTTYPE) )  B1_inst ( B1 , RSTB , CLK , CEB , B1_reg );
assign BCOUT = B1_reg ;

pip_mux  #( .n(18) , .PIPLINING(A1REG) , .RST_TYPE(RSTTYPE) )  A1_inst ( A0_reg , RSTA , CLK , CEA ,
A1_reg );
```

```
assign multiplier_out = B1_reg * A1_reg ;

pip_mux  #( .n(36) , .PIPLINING(MREG) , .RST_TYPE(RSTTYPE) )  M_inst ( multiplier_out , RSTM , CLK , CEM ,
M );

assign cin = CARRYINSEL == "OPMODE5" ? OPMODE[5] : CARRYINSEL == "CARRYIN" ? CARRYIN : 1'b0 ;

pip_mux  #( .n(1) , .PIPLINING(CARRYINREG) , .RST_TYPE(RSTTYPE) )  CIN_inst ( cin , RSTCARRYIN , CLK ,
CECARRYIN , CYI );

assign Z_out = OPMODE[3] ? OPMODE[2] ? 48'b0 : PCIN : OPMODE[2] ? P : C_reg ;
assign X_out = OPMODE[1] ? OPMODE[0] ? 48'b0 : {{12{1'b0}}, M} : OPMODE[0] ? P : {D_reg[11:0], A0_reg,
B0_reg} ;

assign Post_addsub_out = OPMODE[7] ? ({1'b0, Z_out} - {1'b0, X_out} - CYI)
                                   : ({1'b0, Z_out} + {1'b0, X_out} + CYI);

pip_mux  #( .n(48) , .PIPLINING(PREG) , .RST_TYPE(RSTTYPE) )  P_inst ( Post_addsub_out[47:0] , RSTP , CLK
, CEP , P );

pip_mux  #( .n(1) , .PIPLINING(CARRYOUTREG) , .RST_TYPE(RSTTYPE) )  CYO_inst ( Post_addsub_out[48] ,
RSTCARRYIN , CLK , CECARRYIN , CARRYOUT );

assign CARRYOUTF = CARRYOUT ;

endmodule
```

1.2 pip_mux.v

```
module pip_mux ( in, RST, clk,ce , out );
    parameter n = 18;
    parameter PIPLINING = 1;
    parameter RST_TYPE = "SYNC";

    input  [n-1:0] in;
    input  clk, RST , ce;
    output [n-1:0] out;

    reg  [n-1:0] out_reg1, out_reg2;
    wire [n-1:0] reg_out;

    generate

    if (RST_TYPE == "SYNC") begin : sync_gen
        always @(posedge clk) begin
            if (RST)
                out_reg1 <= 0;
            else if (ce)
                out_reg1 <= in;
        end
        assign reg_out = out_reg1;
    end

    else begin : async_gen
        always @(posedge clk or posedge RST) begin
            if (RST)
                out_reg2 <= 0;
            else if (ce)
                out_reg2 <= in;
        end
        assign reg_out = out_reg2;
    end

    endgenerate

    assign out = PIPLINING ? reg_out : in;

endmodule
```

2. Testbench Code

2.1 DSP_tb.v

```
module DSP_tb ;

reg  [17:0] A , B , D , BCIN ;
reg  [47:0] C ;
reg  [7:0] OPMODE ;
reg  CLK, CARRYIN, RSTA, RSTB, RSTM, RSTP, RSTC, RSTD, RSTCARRYIN, RSTOPMODE, CEA, CEB, CEM, CEP, CEC,
CED, CECARRYIN, CEOPMODE, PCIN ;

wire [17:0] BCOUT ;
wire [47:0] PCOUT , P ;
wire [35:0] M ;
wire CARRYOUT , CARRYOUTF ;


DSP DUT (A, B, D, C, BCIN, CLK, CARRYIN, OPMODE, RSTA, RSTB, RSTM, RSTP, RSTC, RSTD, RSTCARRYIN,
RSTOPMODE, CEA, CEB, CEM, CEP, CEC, CED, CECARRYIN, CEOPMODE, PCIN, BCOUT, PCOUT, P, M, CARRYOUT,
CARRYOUTF) ;


initial begin
    CLK = 0;
    forever begin
        #5 CLK = ~CLK ;
    end
end


initial begin

    {RSTA, RSTB, RSTM, RSTP, RSTC, RSTD, RSTCARRYIN, RSTOPMODE} = 8'hFF;
    A = $random;
    B = $random;
    D = $random;
    C = $random;
    BCIN = $random;
    OPMODE = $random;
    CARRYIN = $random;
    PCIN = $random;
    CEA = $random;
    CEB = $random;
    CEM = $random;
    CEP = $random;
    CEC = $random;
    CED = $random;
    CECARRYIN = $random;
    CEOPMODE = $random;

    @(negedge CLK) ;
    if ( BCOUT || PCOUT || P || M || CARRYOUT || CARRYOUTF ) begin
        $display ("ERROR in RST") ;
        $stop ;
    end

    @(negedge CLK) ;
    RSTA = 1'b0;
    RSTB = 1'b0;
    RSTM = 1'b0;
    RSTP = 1'b0;
    RSTC = 1'b0;
    RSTD = 1'b0;
```

```
    RSTCARRYIN = 1'b0;
    RSTOPMODE = 1'b0;

    CEA = 1'b1;
    CEB = 1'b1;
    CEM = 1'b1;
    CEP = 1'b1;
    CEC = 1'b1;
    CED = 1'b1;
    CECARRYIN = 1'b1;
    CEOPMODE = 1'b1;


    OPMODE = 8'b11011101 ;
    A = 20 ; B = 10 ; C = 350 ; D = 25;
    BCIN = $random ; PCIN = $random ; CARRYIN = $random ;
    repeat (4) @(negedge CLK) ;

    if(BCOUT == 'hf && M == 'h12c && P == 'h32 && PCOUT == 'h32 && CARRYOUT == 'b0 && CARRYOUTF == 'b0)
        $display ("Path 1 succeded") ;
    else begin
        $display("ERROR in path 1 ") ;
        $stop ;
    end


    OPMODE = 8'b00010000;
    repeat(3) @(negedge CLK ) ;

    if(BCOUT == 'h23 && M == 'h2bc && P == 'b0 && PCOUT == 'b0 && CARRYOUT == 'b0 && CARRYOUTF == 'b0)
        $display ("Path 2 succeded") ;
    else begin
        $display("ERROR in path 2 ") ;
        $stop ;
    end


    OPMODE = 8'b00001010 ;
    repeat(3) @(negedge CLK ) ;

    if (BCOUT == 'ha &&  M == 'hc8 && P == 'b0 && PCOUT == 'b0 && CARRYOUT == 'b0 && CARRYOUTF == 'b0)
        $display ("Path 3 succeded") ;
    else begin
        $display("ERROR in path 3 ") ;
        $stop ;
    end


    OPMODE = 8'b10100111 ;
    A = 20 ; B = 10 ; C = 350 ; D = 25;
    BCIN = $random  ; CARRYIN = $random ;
    repeat (3) @(negedge CLK) ;

    if(BCOUT == 'h6 && M == 'h1e && P == 'hfe6fffec0bb1 && PCOUT == 'hfe6fffec0bb1 && CARRYOUT == 1 &&
CARRYOUTF == 1)
        $display ("Path 4 succeded") ;
    else begin
        $display("ERROR in path 4 ") ;
        $stop ;
    end

end
endmodule
```

3. Do File

```
vlib work
vmap work work

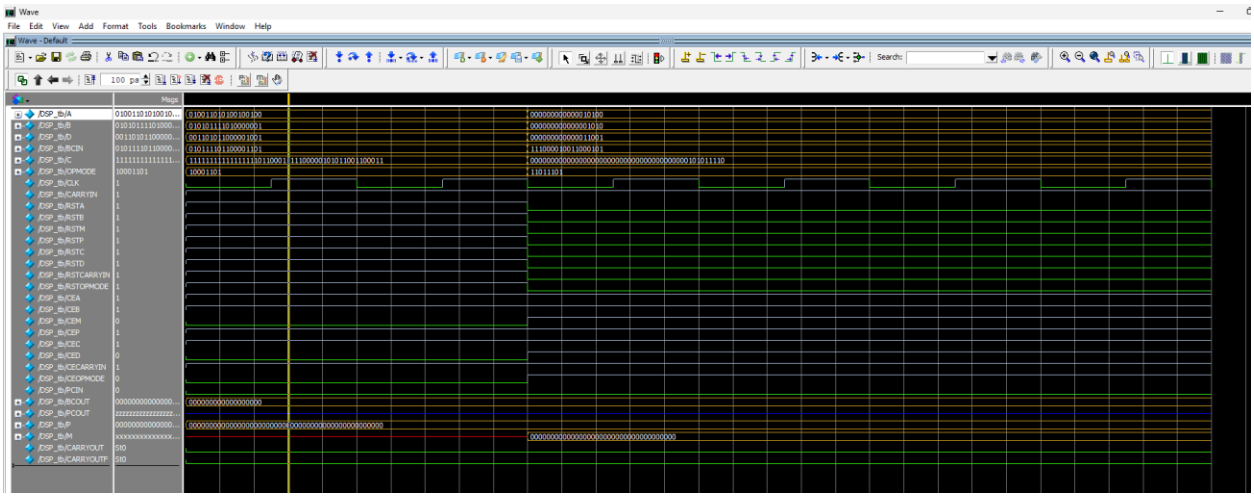
vlog -work work pip_mux.v
vlog -work work DSP.v
vlog -work work DSP_tb.v

vsim -voptargs=+acc work.DSP_tb

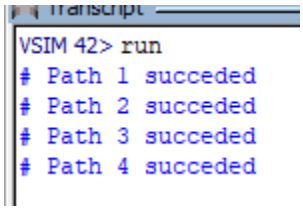
add wave -r sim:/DSP_tb/DUT/*
run -all
```

4. QuestaSim Snippets

Wave window showing DSP_tb signals during simulation.



Transcript window confirming all four test paths succeeded.

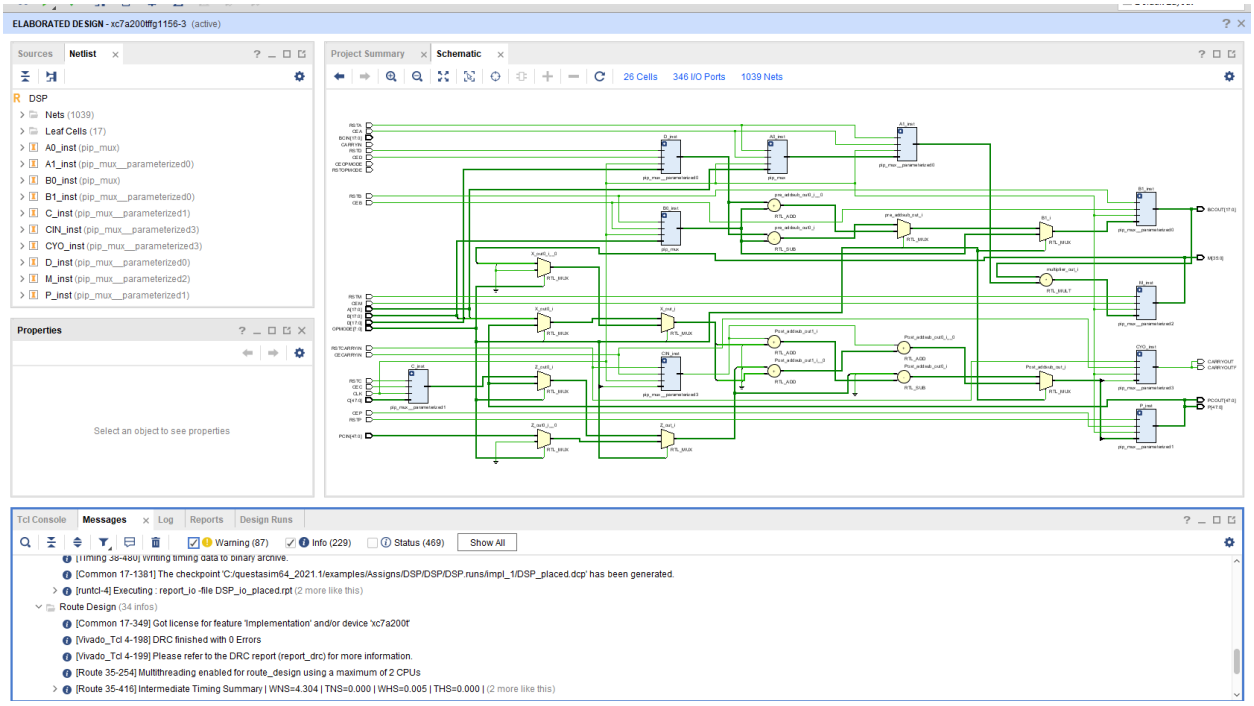


5. Constraint File

```
create_clock -period 10.000 -name CLK -waveform {0.000 5.000} [get_ports CLK]
set_property PACKAGE_PIN W5 [get_ports CLK]
set_property IOSTANDARD LVCMOS33 [get_ports CLK]
```

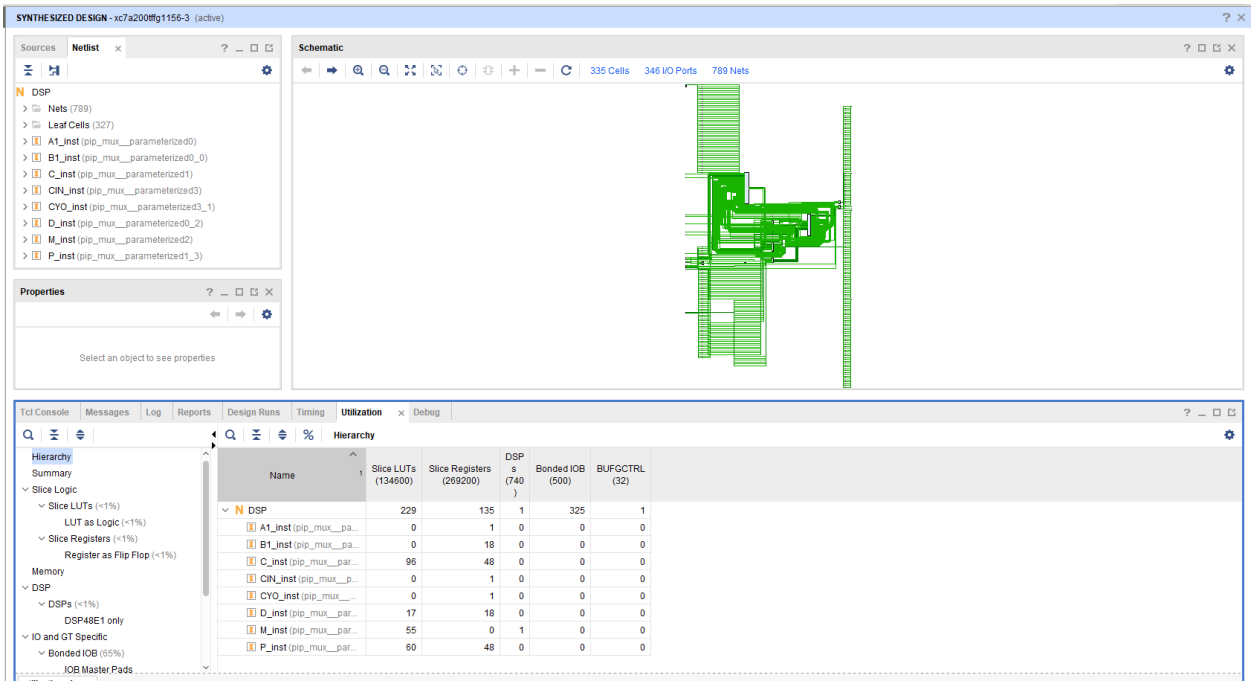
6. Elaboration

Elaborated schematic (DSP block with the pip_mux leaf cells) and the Messages tab after running elaboration.

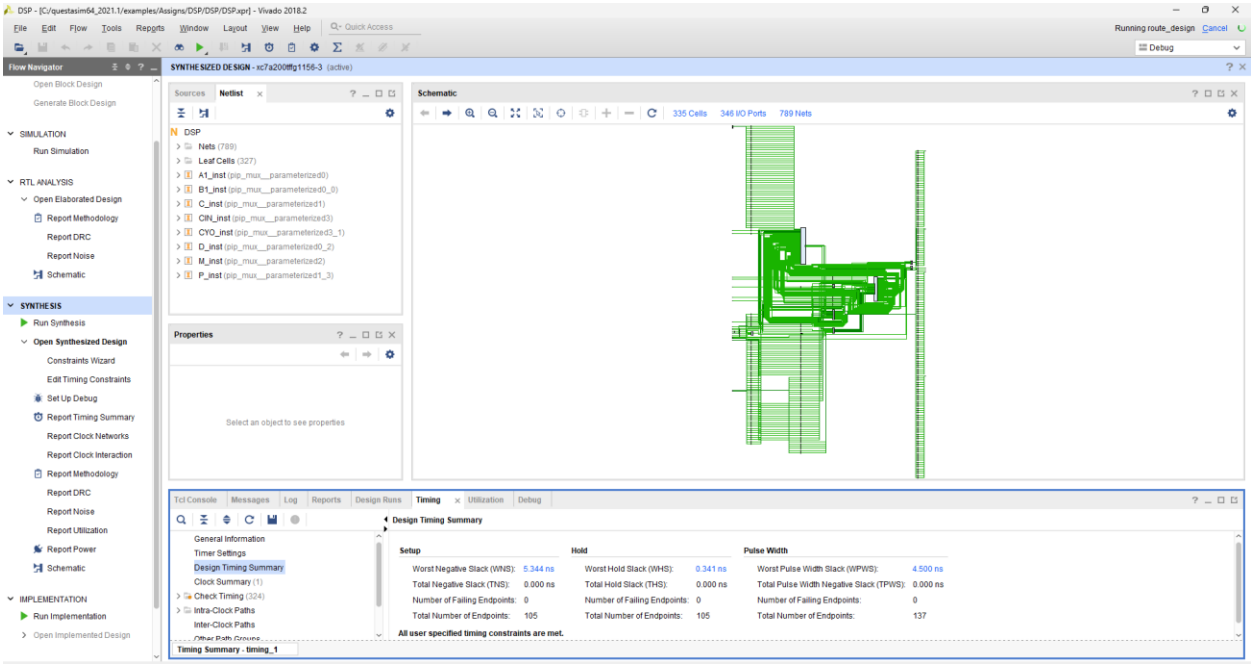


7. Synthesis

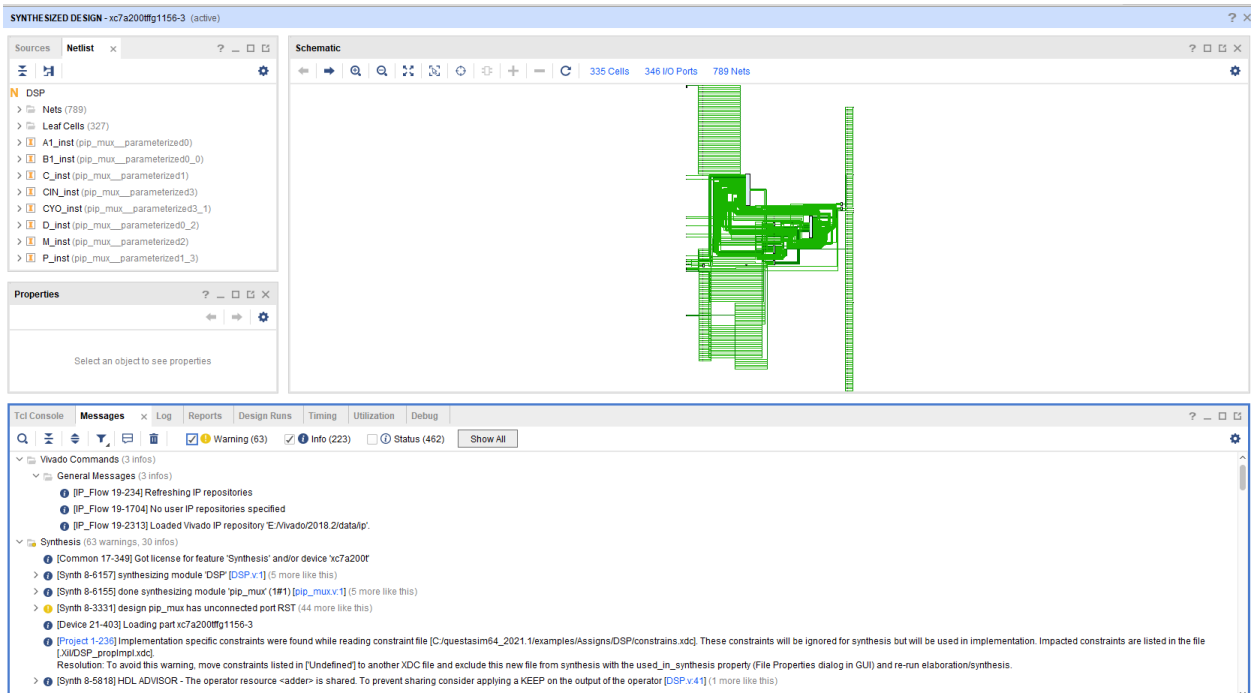
7.1 Schematic and Utilization



7.2 Timing Report

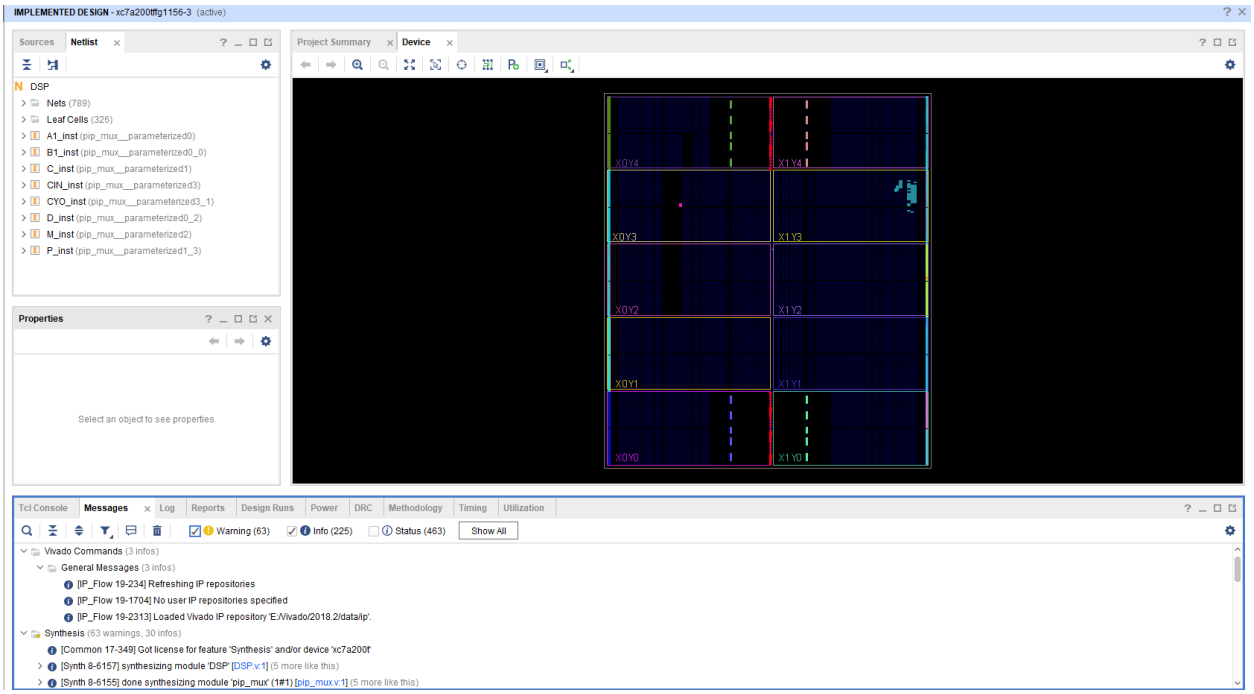


7.3 Messages

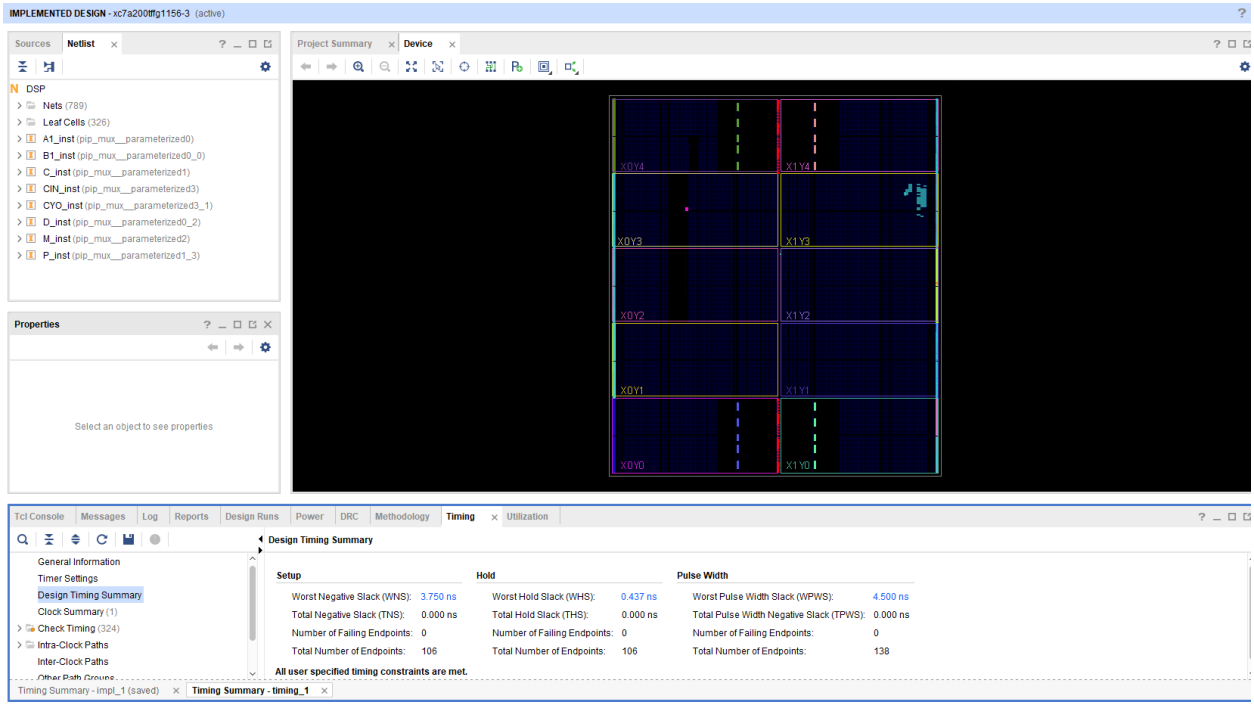

SYNTHESIZED DESIGN - xc7a200tffg1156-3 (active)
Sources
Netlist
DSP
Nets (789)
Leaf Cells (327)
A1_inst (pip_mux__parameterized0)
B1_inst (pip_mux__parameterized0_0)
C_inst (pip_mux__parameterized1)
CIN_inst (pip_mux__parameterized3)
CYO_inst (pip_mux__parameterized3_1)
D_inst (pip_mux__parameterized0_2)
M_inst (pip_mux__parameterized2)
P_inst (pip_mux__parameterized1_3)
Properties
Select an object to see properties
Schematic
335 Cells
346 I/O Ports
789 Nets
Tcl Console
Messages
Log
Reports
Design Runs
Timing
Utilization
Debug
Warning (63)
Info (223)
Status (462)
Show All
Vivado Commands (3 infos)
General Messages (3 infos)
[IP_Flow 19-234] Refreshing IP repositories
[IP_Flow 19-1704] No user IP repositories specified
[IP_Flow 19-2313] Loaded Vivado IP repository 'E:/Vivado/2018.2/data/ip'.
Synthesis (63 warnings, 30 infos)
[Common 17-349] Got license for feature 'Synthesis' and/or device 'xc7a200t'
[Synth 8-6157] synthesizing module 'DSP' [DSP.v:1] (5 more like this)
[Synth 8-6155] done synthesizing module 'pip_mux' (1#1) [pip_mux.v:1] (5 more like this)
[Synth 8-3331] design pip_mux has unconnected port RST (44 more like this)
[Device 21-403] Loading part xc7a200tffg1156-3
[Project 1-236] Implementation specific constraints were found while reading constraint file [C:/questasim64_2021.1/examples/Assigns/DSP/constrains.xdc]. These constraints will be ignored for synthesis but will be used in implementation. Impacted constraints are listed in the file [.Xil/DSP_propImpl.xdc].
Resolution: To avoid this warning, move constraints listed in ['Undefined'] to another XDC file and exclude this new file from synthesis with the used_in_synthesis property (File Properties dialog in GUI) and re-run elaboration/synthesis.
[Synth 8-5818] HDL ADVISOR - The operator resource <adder> is shared. To prevent sharing consider applying a KEEP on the output of the operator [DSP.v:41] (1 more like this)

8. Implementation

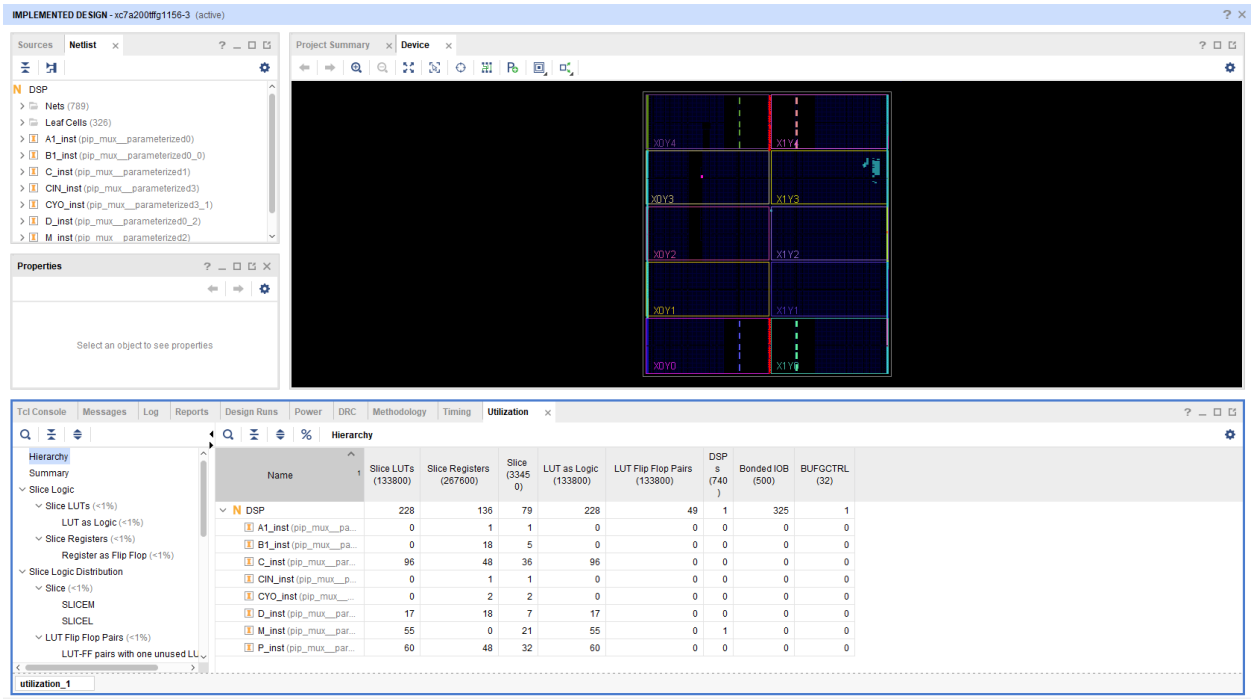
8.1 Device View, Netlist and Messages



8.2 Timing Report

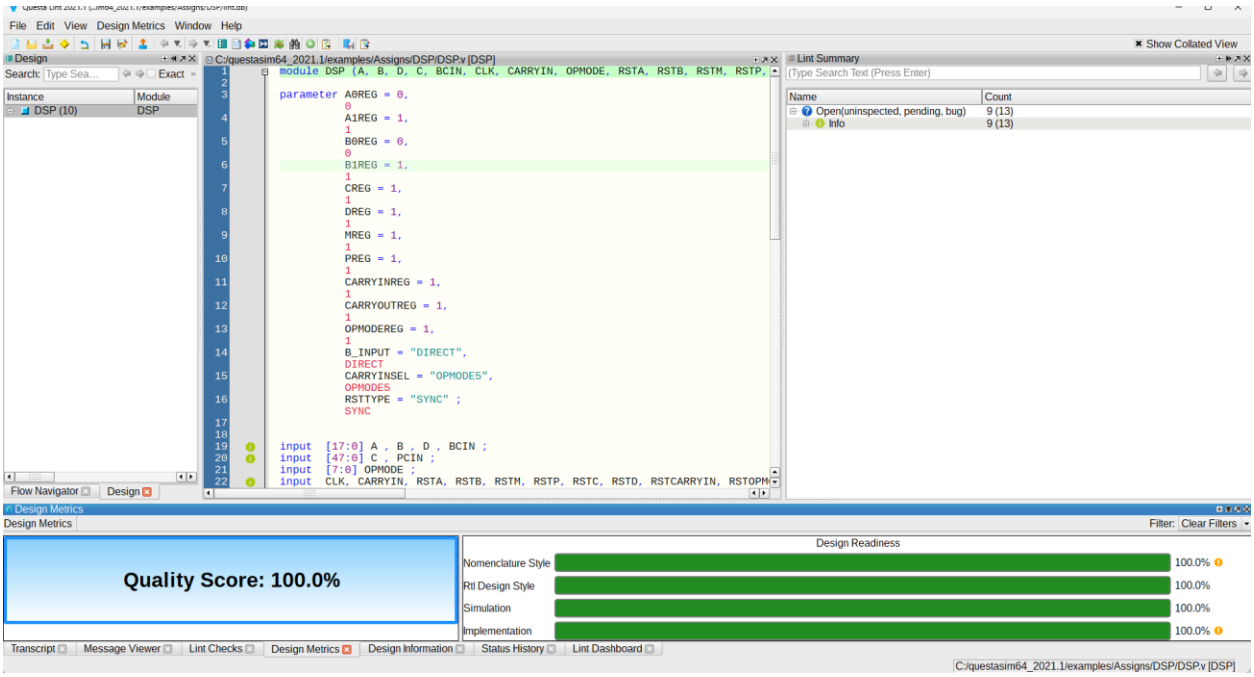


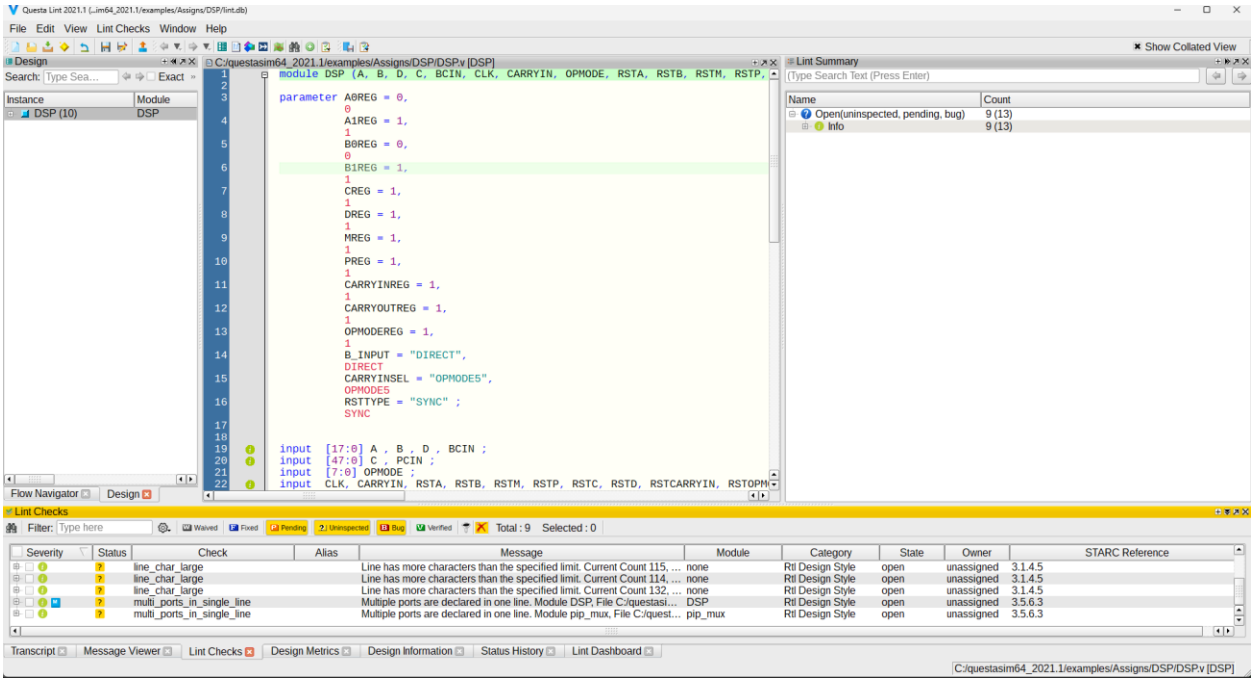
8.3 Utilization Report



Name	Slice LUTs (133800)	Slice Registers (267600)	Slice (33450)	LUT as Logic (133800)	LUT Flip Flop Pairs (133800)	DSPs (740)	Bonded IOB (500)	BUFGCTRL (32)
DSP	228	136	79	228	49	1	325	1
A1_inst (pip_mux__pa...	0	1	1	0	0	0	0	0
B1_inst (pip_mux__pa...	0	18	5	0	0	0	0	0
C_inst (pip_mux__par...	96	48	36	96	0	0	0	0
CIN_inst (pip_mux__p...	0	1	1	0	0	0	0	0
CYO_inst (pip_mux__...	0	2	2	0	0	0	0	0
D_inst (pip_mux__par...	17	18	7	17	0	0	0	0
M_inst (pip_mux__par...	55	0	21	55	0	1	0	0
P_inst (pip_mux__par...	60	48	32	60	0	0	0	0

9. Linting


File Edit View Design Metrics Window Help
Show Collated View
Design
Search: Type Sea... Exact
Instance Module
DSP (10) DSP
C:/questasim64_2021.1/examples/Assigns/DSP/DSP.v [DSP]
module DSP (A, B, D, C, BCIN, CLK, CARRYIN, OPMODE, RSTA, RSTB, RSTM, RSTP,
parameter A0REG = 0,
A1REG = 1,
B0REG = 0,
B1REG = 1,
CREG = 1,
DREG = 1,
MREG = 1,
PREG = 1,
CARRYINREG = 1,
CARRYOUTREG = 1,
OPMODEREG = 1,
B_INPUT = "DIRECT",
CARRYINSEL = "OPMODE5",
RSTTYPE = "SYNC" ;
input [17:0] A , B , D , BCIN ;
input [47:0] C , PCIN ;
input [7:0] OPMODE ;
input CLK, CARRYIN, RSTA, RSTB, RSTM, RSTP, RSTC, RSTD, RSTCARRYIN, RSTOPM
Lint Summary
(Type Search Text (Press Enter))
Name Count
Open(uninspected, pending, bug) 9 (13)
Info 9 (13)
Flow Navigator Design
Design Metrics
Design Metrics
Filter: Clear Filters
Quality Score: 100.0%
Design Readiness
Nomenclature Style 100.0%
Rtl Design Style 100.0%
Simulation 100.0%
Implementation 100.0%
Transcript Message Viewer Lint Checks Design Metrics Design Information Status History Lint Dashboard
C:/questasim64_2021.1/examples/Assigns/DSP/DSP.v [DSP]


File Edit View Lint Checks Window Help
Show Collated View
Design
Search: Type Sea... Exact
Instance Module
DSP (10) DSP
C:/questasim64_2021.1/examples/Assigns/DSP/DSP.v [DSP]
module DSP (A, B, D, C, BCIN, CLK, CARRYIN, OPMODE, RSTA, RSTB, RSTM, RSTP,
parameter A0REG = 0,
A1REG = 1,
B0REG = 0,
B1REG = 1,
CREG = 1,
DREG = 1,
MREG = 1,
PREG = 1,
CARRYINREG = 1,
CARRYOUTREG = 1,
OPMODEREG = 1,
B_INPUT = "DIRECT",
CARRYINSEL = "OPMODE5",
RSTTYPE = "SYNC" ;
input [17:0] A , B , D , BCIN ;
input [47:0] C , PCIN ;
input [7:0] OPMODE ;
input CLK, CARRYIN, RSTA, RSTB, RSTM, RSTP, RSTC, RSTD, RSTCARRYIN, RSTOPM
Lint Summary
(Type Search Text (Press Enter))
Name Count
Open(uninspected, pending, bug) 9 (13)
Info 9 (13)
Flow Navigator Design
Lint Checks
Filter: Type here
Waived Fixed Pending Uninspected Bug Verified
Total : 9 Selected : 0
Severity Status Check Alias Message Module Category State Owner STARC Reference
line_char_large Line has more characters than the specified limit. Current Count 115, ... none Rtl Design Style open unassigned 3.1.4.5
line_char_large Line has more characters than the specified limit. Current Count 114, ... none Rtl Design Style open unassigned 3.1.4.5
line_char_large Line has more characters than the specified limit. Current Count 132, ... none Rtl Design Style open unassigned 3.1.4.5
multi_ports_in_single_line Multiple ports are declared in one line. Module DSP, File C:/questasi... DSP Rtl Design Style open unassigned 3.5.6.3
multi_ports_in_single_line Multiple ports are declared in one line. Module pip_mux, File C:/quest... pip_mux Rtl Design Style open unassigned 3.5.6.3
Transcript Message Viewer Lint Checks Design Metrics Design Information Status History Lint Dashboard
C:/questasim64_2021.1/examples/Assigns/DSP/DSP.v [DSP]